Reproducibility Report

Project 2: Longitudinal CEA Trajectories and Disease Progression Prediction in Colorectal Cancer

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This report documents the reproducibility practices used for Project 2. It is organized around the five practice areas covered in the course's reproducibility lecture: file organization and naming, version control, safe programming, dynamic reports, and pipelines.

Project links.

- Code repository (GitHub): https://github.com/yananf8899/Biostat629
- Project website (GitHub Pages): https://yananf8899.github.io/Biostat629/
- Manuscript PDF (in repository): `project2_report.pdf`

1 File Organization and Naming

The project follows a flat layout with descriptive, lowercase, underscore-separated file names. The lecture's recommendation to date-stamp working files is applied to interim notes; the final deliverables use stable names so that external links (e.g., the website's `report.pdf`) do not break across edits.

```
.
|-- README.md                  project-level reproducibility notes
|-- Project2_Proposal.md       original proposal
|-- project2_analysis.R        single-script analysis pipeline
|-- project2_report.tex/.pdf   final manuscript
|-- project2_presentation.pptx in-class slides
|-- project2_poster.pptx       poster
|-- reproducibility_report.tex/.pdf  this document
|-- sessionInfo.txt            package versions captured at run time
|-- fig{1..5}_*.pdf / .png     figures generated by the R script
|-- docs/                      GitHub Pages website
|   |-- index.html
|   |-- style.css
|   |-- figures/
|   `-- report.pdf
`-- msk_chord_2024/            controlled-access data (NOT in git)
```

The raw input directory `msk_chord_2024/` is referenced by a single relative path variable inside the R script (`data_dir <- "msk_chord_2024"`), and the directory itself is excluded from version control via `.gitignore` because MSK-CHORD is controlled-access data that may not be redistributed.

2 Version Control

The project is tracked with `git` and hosted on GitHub at https://github.com/yananf8899/Biostat629. The repository contains every file needed to reproduce the manuscript, the website, and the figures, except the controlled-access input data. The git history records the development state of the analysis and manuscript and allows the project to be versioned and shared transparently.

A minimal `.gitignore` excludes the raw data folder and intermediate LaTeX build artifacts:

```
msk_chord_2024/
*.aux
*.log
*.out
.DS_Store
```

3 Safe Programming Practices

The analysis script `project2_analysis.R` follows the practices recommended in lecture:

- **Raw data is never modified in place.** All filtering, reshaping, and feature construction happens in R memory using `data.table`; the files in `msk_chord_2024/` are read but never written.
- **No use of `setwd()`, `rm()`, `source()`, or `system()`.** The data path is a single top-of-file variable (`data_dir`); the script does not change the working directory or call out to the shell.
- **No interactive `save.image()` / `load(".RData")`.** Each run starts from a clean session.
- **Random seeds are fixed.** `set.seed(629)` is called immediately before each random step: the patient-level 70/30 train/test split, the LASSO 5-fold cross-validation foldid, the conformal calibration sub-split, and the figure-1 patient sample. Every reported number reproduces exactly.
- **No hard-coded results.** All numbers cited in the manuscript and on the website (AUCs, p-values, calibration slopes, conformal coverage, death-in-window rates) are printed by the script; no value is typed by hand.
- **Numerical guards.** CEA values are clamped to non-negative before `log()` via the offset `log(RESULT + 0.1)`; predicted probabilities are clamped to $[10^{-6}, 1 - 10^{-6}]$ before being passed to `qlogis()` for the calibration fit.
- **Patient-level integrity.** Train/test and CV folds are formed at the *patient* level (not the landmark level), so no patient appears in both train and test, and the LASSO CV foldid uses a per-patient lookup so all of a patient's landmarks share a fold.
- **Diagnostic and audit blocks.** The script prints (i) cohort attrition counts, (ii) the death-in-window competing-risk rate, and (iii) a high-precision Model A vs. Model B AUC audit (six-decimal AUCs and rank correlation), so reviewers can verify the headline numbers without re-running ad hoc analyses.

4 Dynamic Reports / Environment Capture

The manuscript is written in plain LaTeX (project2_report.tex) rather than R Markdown. The lecture recommends dynamic reports because they tie text and code together, but the project uses a workable equivalent: a single deterministic R script writes every figure to disk under a fixed filename, and the LaTeX document includes those filenames. Re-running the R script and recompiling pdflatex regenerates the manuscript end-to-end.

To capture the software environment, the script ends with:

```
capture.output(sessionInfo(), file = "sessionInfo.txt")
```

so that R's version, the loaded packages, and their versions for the most recent run are committed to the repository as sessionInfo.txt. Anyone re-running the analysis can compare against this file to detect environment drift.

5 Reproducible Pipeline

The project does not require a Snakemake or Docker pipeline because the analysis is small enough to fit in a single R script. The pipeline is deliberately one command:

```
Rscript project2_analysis.R          # generates fig1..fig5 and sessionInfo.txt
pdflatex project2_report.tex         # compiles the manuscript (twice for refs)
pdflatex project2_report.tex
pdflatex reproducibility_report.tex  # compiles this document
```

Running these four commands on a fresh checkout (with msk_chord_2024/ placed at the project root) regenerates every figure, every table, every number cited in the manuscript and on the website, and the manuscript and reproducibility-report PDFs. Run time is approximately three minutes on a 2023 MacBook Pro, dominated by the glmer() fit for Model A.

Software requirements.

- R $\geq$ 4.3.0
- R packages: data.table, lme4, pROC, ggplot2, glmnet
- TeX Live with pdflatex

Exact versions are captured in sessionInfo.txt.

6 Data Access

MSK-CHORD 2024 is controlled-access. Reviewers must obtain the release through cBioPortal (https://cbioportal.org) and unpack it into a folder named msk_chord_2024/ at the project root. The script expects the following seven files inside that folder:

```
data_clinical_patient.txt
data_clinical_sample.txt
data_timeline_cea_labs.txt
data_timeline_progression.txt
data_timeline_diagnosis.txt
data_timeline_treatment.txt
data_mutations.txt
```

No data are committed to the repository, and the script does not modify any of these files.

7 Summary Checklist

Practice	Status
Descriptive file names, flat directory layout	yes
Project tracked in `git`, hosted on GitHub	yes (link above)
`.gitignore` excludes controlled-access data and build artifacts	yes
Raw data never modified in place	yes
No `setwd`, `rm`, `source`, `system` in scripts	yes
Random seeds fixed for every stochastic step	yes (`set.seed(629)`)
Patient-level train/test and CV folds (no leakage)	yes
All cited numbers printed by the script (no hand-typed values)	yes
Software environment captured in `sessionInfo.txt`	yes
Single-command reproduction	yes (4-line shell pipeline)
Project website with link to repo and report	yes (link above)